

Electric circuits virtual labs

Year 10 GCSE Physics • Three investigations of charge, energy and resistance

Name _____

Partner _____

Predict. Test. Explain.

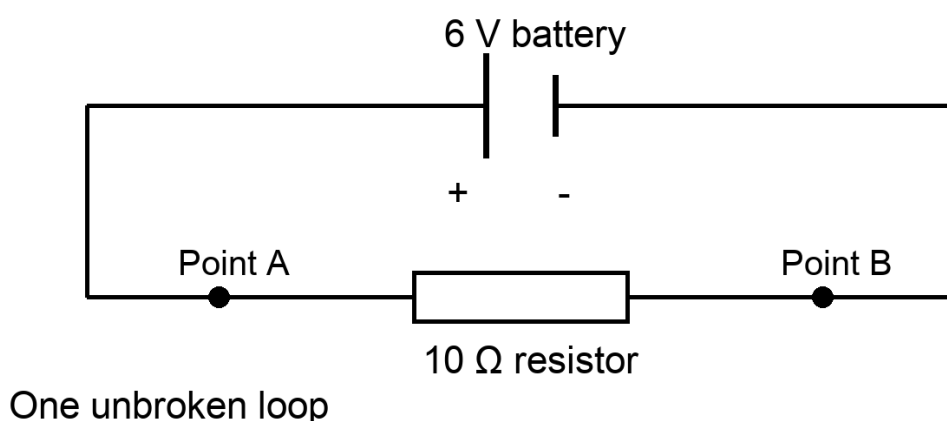
Work in pairs. Swap roles on each new page.

[Open PhET Circuit Construction Kit DC — Lab](#)

On paper: search “PhET Circuit Construction Kit DC”. Choose Lab.

Set up once

1. Select Lab. Turn on Values and Show Current.
Start with Electrons.
2. Build the loop below. Join the wire ends together.
3. Tap the battery: set 6 V. Tap the resistor: set 10 Ω .
4. In Advanced Options, set Wire Resistivity and Battery Resistance to their minimum settings.



Meters and key words

Measure current

Drag out the Non-Contact Ammeter. Place its probe on a wire. You do not need to break the circuit.

A normal wired ammeter goes in series.

Measure potential difference

Drag out a voltmeter. Put one probe at each end of the component. Swap the probes if the reading is negative.

Keep these ideas separate

Quantity	Meaning	Unit
Charge Q	An amount of electric charge	coulomb (C)
Current I	Charge passing a point each second	ampere (A)
Potential difference V	Energy transferred per coulomb between two points	volt (V)
Energy E	Energy transferred electrically	joule (J)
Resistance R	Ratio of p.d. to current for a component	ohm (Ω)

The moving dots show flow. One dot is not one coulomb. Use the meters. Do not count dots or time their journey.

Lab 1 Is current used up

Lab 1 • Current and charge • $I = Q/t$

1 Predict before you measure

Mia says: “The resistor uses up some current. There will be less at B than at A.”

Do you agree? Explain before measuring.

2 Test the claim

Keep the battery at 6 V and the resistor at 10 Ω . Measure at A, at B and on a third wire near the battery. Move only the ammeter probe.

Position	Current I / A
A	
B	
Third position	

What do your readings show about current in this single loop? Use numbers in your answer.

Lab 1 Break and restore the circuit

Open a connection: select a joint and use its cut control (or Delete). Then reconnect it.

What happens to the current when the loop is broken?

Lab 1 Explain the flow

Switch between Electrons and Conventional. In the wires outside the battery, complete:

Electrons move from the _____ terminal towards the _____ terminal.

Conventional current points from _____ towards _____.

Does switching the display change the size of the measured current?

4 Explain it to a younger pupil

Rewrite Mia's claim using the words charge and energy.

Support: "The charge continues around the circuit. In the resistor, energy ..."

Lab 1 How much charge passes

Continue the same circuit • A current reading describes a rate

Current is charge passing a point per second. For steady current:

$$I = Q \div t$$

$$Q = I \times t \qquad t = Q \div I$$

I in amperes (A), Q in coulombs (C), t in seconds (s). One ampere means one coulomb per second.

5 From a meter reading to an amount

Example: if $I = 0.60 \text{ A}$, then 0.60 C passes a point each second.

In 5.0 s , $Q = I \times t = 0.60 \times 5.0 = 3.0 \text{ C}$.

Measure I with a 10Ω resistor at each battery setting. Calculate the charge passing one point in 10 s and 20 s . The times are given intervals: no stopwatch is needed.

Battery setting / V	Current / A	Q in 10 s / C	Q in 20 s / C
3			
6			
9			

Show one charge calculation, including the unit.

Lab 1 Think about the rate

At the same current, doubling the time _____ the charge transferred.
For the same time, doubling the current _____ the charge transferred.

If you wait for twice as long, does the current itself have to double? Explain.

Lab 1 Check your understanding

a. A charge of 12 C passes a point in 20 s. Calculate the current.

b. At 0.30 A, how long does it take for 9.0 C to pass a point?

c. A student says “0.60 A means there is 0.60 C in the whole circuit.” Correct this.

Lab 2 What does a volt mean

Lab 2 • Energy per coulomb • $V = E/Q$

Potential difference tells us how much energy is transferred per coulomb. A p.d. of 6 V across a resistor means 6 J is transferred electrically to the resistor for every 1 C passing through it.

$$V = E \div Q$$

$$E = V \times Q$$

$$Q = E \div V$$

V in volts (V), E in joules (J), Q in coulombs (C). One volt means one joule per coulomb.

1 Predict and measure

Predict: for the same 2 C of charge, will more energy be transferred at 3 V or 9 V? Why?

Use one 10 Ω resistor. At each battery setting, measure the p.d. across the battery, then across the resistor. Touch the two voltmeter probes to opposite ends of the component. Swap probes if the reading is negative.

Battery setting / V	Battery p.d. / V	Resistor p.d. / V	E for 2 C / J
3			
6			
9			

Energy is calculated here, not measured by PhET. Use $E = V \times Q$ and your measured resistor p.d. Small differences from the battery setting are possible.

Lab 2 Explain potential difference

Show the calculation for the 6 V setting.

At 9 V, each coulomb transfers _____ J in a resistor with 9 V across it.
At 3 V, each coulomb transfers _____ J. The amount of charge in this comparison is the same; the _____ per coulomb changes.

A resistor transfers 24 J when 4 C passes through it. Calculate its p.d.

3 A p.d. needs two points

At 6 V, place both probes on the same piece of wire, then on opposite ends of the resistor. Readings: same wire _____ V; resistor _____ V.
Why is “the voltage at the resistor” less precise than “the voltage across it”?

Lab 2 Follow the energy

Continue Lab 2 • Charge is conserved while energy is transferred

4 Share the energy between two resistors

Set the battery to 6 V. Add a second 10 Ω resistor in series, so there is still only one loop. Label the resistors R1 and R2. Predict the p.d. across each.

My prediction: R1 _____ V; R2 _____ V. Reason:

Measure the battery p.d. and the p.d. across each resistor. Also measure current on either side of R1.

Battery p.d. / V	R1 p.d. / V	R2 p.d. / V

Current before R1 _____ A; current after R1 _____ A.

Lab 2 Account for the energy

Use your voltage readings from the previous page.

Complete the energy account for 2 C of charge.
Ignore the tiny energy transfer in the wires.

Where energy is transferred	Calculation $E = V \times Q$	Energy / J
Supplied by the battery		
Transferred to R1		
Transferred to R2		

Complete: the battery supplies about _____ J; the two resistors together receive about _____ J. This supports conservation of _____.

Why can the current be the same before and after a resistor even though energy is transferred there?

Lab 2 Compare two circuits

Return to one resistor. Measure the current and resistor p.d. in each circuit below. Use the measured p.d. to calculate energy transferred per coulomb.

Circuit settings	Current / A	Measured p.d. / V	Energy per C / J
A: 6 V battery, 10 Ω			
B: 12 V battery, 20 Ω			

Compare charge passing per second and energy transferred per coulomb in A and B. Explain why current and p.d. describe different things.

Lab 3 Discover the resistance rule

Lab 3 • Current, potential difference and resistance

1 Predict and collect evidence

Set $R = 10\ \Omega$. Predict: if the p.d. doubles, the current will _____.

Change only the battery setting. Keep the resistor and wires the same. Measure the p.d. across the resistor and the current through the loop at each setting.

Battery setting / V	Resistor p.d. / V	Current / A	Calculate $V \div I / \Omega$
2			
4			
6			
8			
10			
12			

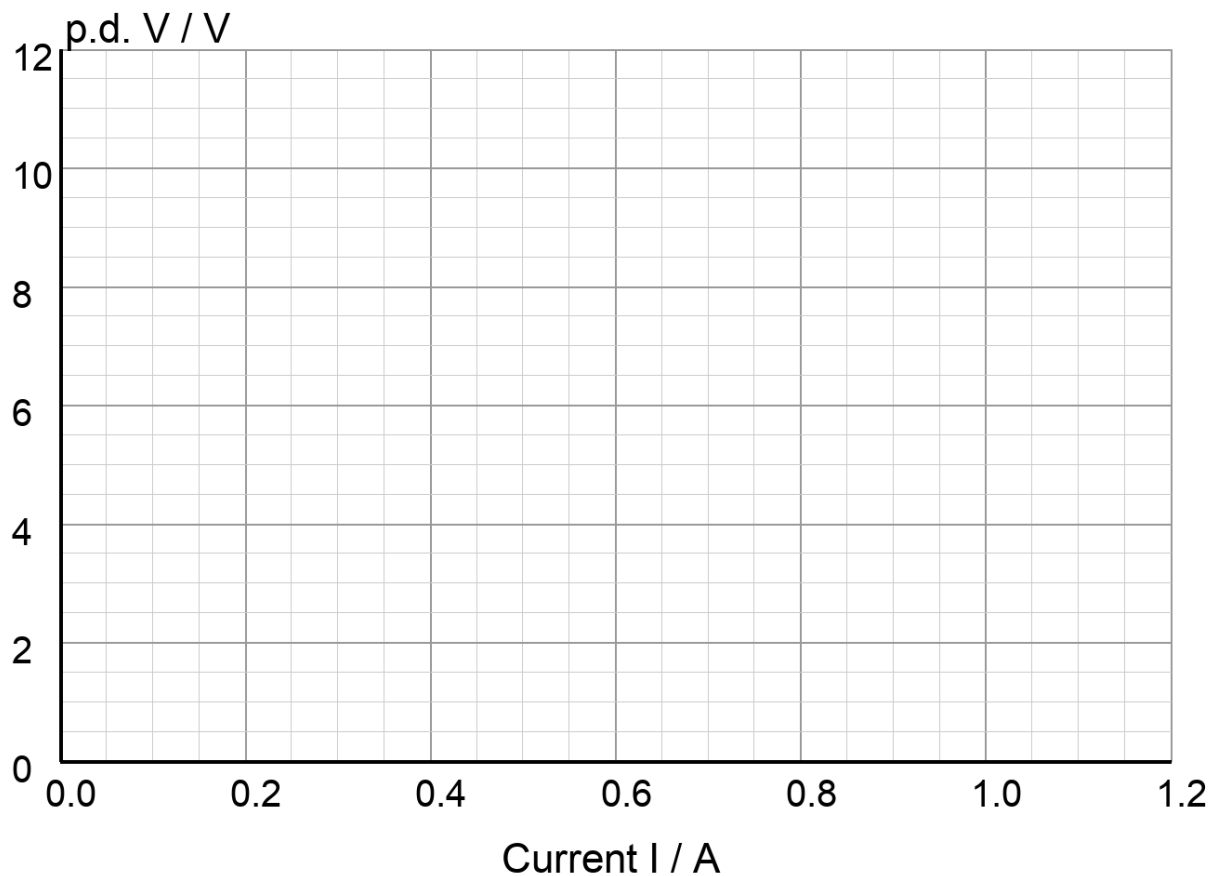
The ratio $V \div I$ is approximately _____ Ω . What component setting does it match?

Lab 3 Plot your results

Plot current on the horizontal axis.

Plot p.d. on the vertical axis.

Draw a straight line of best fit if your points support it.



Describe the line and whether it passes close to the origin. What does it show about V and I when resistance stays constant?

For a fixed resistor at constant temperature: $V = I \times R$.

V in volts (V), I in amperes (A), R in ohms (Ω).

Lab 3 Design a circuit to order

Continue Lab 3 • Use evidence to choose settings before you test them

3 Change resistance

Keep the battery at 6 V. Predict before measuring: if resistance doubles, current will _____. Now test the three resistances below.

Resistance R / Ω	Resistor p.d. / V	Current / A
5		
10		
20		

What pattern do you find? Keep the p.d. approximately constant in your explanation.

Rearrange the rule: $I = V \div R$ and $R = V \div I$.

Example: at 6 V with 20 Ω , $I = 6 \div 20 = 0.30$ A.

Lab 3 The design challenge

Design a single-resistor circuit that transfers 18 J to the resistor while 3.0 C passes through it in 5.0 s. Work out the settings before building.

Find	Equation and working	Answer with unit
Current I		
Resistor p.d. V		
Resistance R		

Build your design. Choose a battery setting close to your required resistor p.d.
Measured current _____ A; measured resistor p.d. _____ V.
Use those readings to check: Q in 5.0 s = _____ C; E = _____ J.

Did your circuit meet the target approximately? Explain any small difference.

Lab 3 Exit ticket

a. What does a current of 0.40 A mean?

b. What does a p.d. of 8 V across a resistor mean?

c. A $15\ \Omega$ resistor has 0.20 A through it. Calculate the p.d. across it.

d. Which is transferred to the surroundings at a resistor: charge or energy?
